

ENGR 095

Fall 2026 - Jenn Winikus & Nick Strandwitz

Diane Ngo

23 September 2026

Sensing the World - Microcontrollers Reflection 2

In this lab module, we dived deeper into the computer engineering side of the module. Firstly, we created Github accounts and using codespace and AI, we were able to make our own portfolio and then set up 2 git repositories, one for our professional personal portfolio and one for our FYRE assignments. Next, we explored the basics of python programming on Arduino with an Arduino Nano ESP32. After learning basic micropython and lighting up a LED on with the arduino, we moved into the Keysight lab in Packard. We learnt more about electronics, things like breadboards, LEDs, pins and ports, motors, humidity sensors, and more. We went through a few small projects to make our own circuits with those suppliers. First we made LEDs light up, rested different voltages and experimented with resistors breaking. We also used a humidity sensor and calculated the data from that and our circuits to test it with automatic code taken from AI.

Through this experience, I learned a lot about circuits. I had some experience before competing in Circuit Lab for Science Olympiad, but this was a lot more different and a lot harder than I expected. I think for me, the more difficult part was knowing which pins and ports to connect the wires and things to on the breadboard. I understood the concept and how the breadboard worked but doing it myself was challenging. I found this unit a lot more engaging and fun.